

Local AI Lab

Five-run on-device model benchmark

Gemini Nano was the decisive performance and representative-output winner: 1.32 s median latency, tight 1.26-1.34 s consistency, and correct placement of the 08:30 medication event.

Executive summary

Local AI Lab ran the same instruction and passage five times through each of three on-device inference paths on a physical Pixel 11 Pro XL. All 15 attempts completed. The full operational suite took 3 minutes 10 seconds, including engine unloading, model loading, session creation, transitions, and inference.

- **Gemini Nano through AICore** delivered the lowest median latency (1.32 s) and the highest estimated end-to-end output rate (77.9 est. tok/s).
- **gemma-2b-it-cpu-int4.bin through MediaPipe** was stable at 4.45 s median latency but its representative output omitted both medications and introduced unsupported details.
- **Llama-3.2-3B-Instruct-Q4_K_M.gguf through llama.cpp** was much slower and more variable, with a 25.90 s median and an 18.30-50.75 s range. Its representative output moved the medication event from 08:30 to 09:30.

Controlled setup

Field	Recorded condition
Device	Google Pixel 11 Pro XL; Android 17
Application	Local AI Lab release build; database version 4
AICore build	0.release.prod_aicore_20260723.00_RC11.964081323 (versionCode 494417)
Repetitions	5 runs per model; 15 total attempts
Model order	Gemini Nano, Gemma 2B, Llama 3.2 3B
Suite duration	3 minutes 10 seconds wall-clock
Network claim	No network-isolation claim was made for this run

Fixed benchmark input

Instruction: Extract all timestamps, vitals, and medication events chronologically:

Passage: Client arrived at 08:30 for morning medication (Metformin 500mg, Lisinopril 10mg) with full glass of water. Refused breakfast initially citing mild nausea, but ate half a banana at 09:15. Blood glucose reading at 09:30 was 128 mg/dL. Attended physical therapy session from 10:00 to 10:45 with good mobility and no complaints of pain. Returned to common area for lunch at 12:00.

Performance results

Medians are calculated across five completed runs. Token counts are estimated from output characters divided by four. Estimated output rate uses total inference latency, including prompt processing and time to first token; it is an end-to-end proxy, not tokenizer output or pure decode speed.

Model / runtime	Runs	Median latency	Range	Median TTFT	Median est. rate	Median est. tokens
Gemini Nano AICore	5/5	1.32 s	1.26-1.34 s	N/A*	77.9	103
gemma-2b-it-cpu-int4.bin MediaPipe	5/5	4.45 s	4.42-4.55 s	0.50 s	18.7	83
Llama-3.2-3B-Instruct-Q4_K_M.gguf llama.cpp	5/5	25.90 s	18.30-50.75 s	9.35 s	4.2	118

* Gemini Nano's Prompt API is non-streaming. True TTFT cannot be measured; the application displays total latency with an asterisk in the TTFT column.

Per-run measurements

Model	Run	Latency (s)	Est. tok/s	Est. tokens
Gemini Nano	1	1.34	77.0	103
	2	1.26	81.6	103
	3	1.30	79.3	103
	4	1.32	76.7	101
	5	1.32	77.9	103
Gemma 2B	1	4.42	18.8	83
	2	4.42	18.8	83
	3	4.47	18.6	83
	4	4.45	18.7	83
	5	4.55	18.2	83
Llama 3.2 3B	1	28.82	4.1	119
	2	23.02	6.2	143
	3	50.75	2.3	118
	4	18.30	5.2	95
	5	25.90	4.2	109

Relative median latency: Gemini Nano was about 3.4x faster than Gemma 2B and 19.6x faster than Llama 3.2 3B. Gemma was about 5.8x faster than Llama.

Representative-output assessment

The application exposes the output from the median-latency run for each model. The assessment below evaluates those three representative outputs, not every saved output from all 15 attempts.

Model	Observed representative-output behavior	Assessment
Gemini Nano	Placed Metformin 500mg and Lisinopril 10mg at 08:30; retained the 09:15 banana, 09:30 glucose value, 10:00-10:45 therapy interval, and 12:00 return for lunch.	Best factual preservation
Gemma 2B	Omitted both medications; changed arrival to 'at the clinic'; treated 10:45 as a return to the common area; changed the 12:00 event to having lunch.	Material omissions and unsupported changes
Llama 3.2 3B	Included the medication names and doses but moved the medication event from 08:30 to 09:30. Other major timestamps were retained.	Material chronological error

Gemini Nano representative output

Here are the extracted events in chronological order:

1. 08:30: Client arrived for morning medication (Metformin 500mg, Lisinopril 10mg) with a full glass of water.
2. 09:15: Ate half a banana.
3. 09:30: Blood glucose reading was 128 mg/dL.
4. 10:00 to 10:45: Attended physical therapy session (good mobility, no complaints of pain).
5. 12:00: Returned to the common area for lunch.

Why quality changes the benchmark verdict

Latency alone would rank the models in the same order, but this prompt has a high-value factual requirement: medications must remain attached to the correct time. Gemma's omission and Llama's reassignment are not cosmetic differences. For caregiver documentation or chronological extraction, either could change the meaning of the source note. Human review remains necessary for every path.

Interpretation and limitations

- This is one fixed passage and instruction on one physical device. It does not establish general model quality across domains.
- Only five runs per model were collected. Medians and ranges describe this session, not long-term production behavior.
- The application benchmarks models sequentially rather than randomizing model order. Loading, caching, and thermal history may therefore affect later runs.
- Displayed inference latency excludes model loading, engine unloading, session creation, and inter-run delays. The 3:10 wall-clock duration includes those operational costs.
- Estimated token counts use output characters divided by four. Each engine has a different tokenizer, so the estimated rates are comparative proxies.
- Gemini Nano is non-streaming through the tested Prompt API, so true time to first token is unavailable.
- The recorded AICore package build was 0.release.prod_aicore_20260723.00_RC11.964081323 (versionCode 494417). This attributes the serving system software, not the precise proprietary Gemini Nano checkpoint or weights.
- The representative-output review covers the median-latency output shown for each model, not every output from all five runs.
- No network-isolation claim is made for this specific benchmark session.

Final verdict

For this controlled chronological-extraction task, Gemini Nano was the clear winner. It combined the lowest and most consistent latency with the strongest representative factual preservation. Gemma 2B was substantially slower and omitted the central medication event. Llama 3.2 3B was slowest, highly variable, and reassigned that event to the wrong timestamp.

Authorship and assistance

Jason McArdle directed the project and expanded its scope from an initial single-model summarizer into Local AI Lab, incorporating multiple inference paths and comparative benchmarking. He determined the product direction and testing decisions, performed the physical-device benchmark, reviewed the recorded outputs, identified factual errors, and accepts responsibility for the findings and publication. Gemini suggested the initial offline summarizer concept and provided substantial implementation and troubleshooting assistance during development. ChatGPT became involved after the core application had been built and assisted later with application refinement, benchmark methodology, persistence improvements, data interpretation, and report preparation. All measurements and model outputs reported here came from Local AI Lab running on physical Pixel hardware; none were simulated or generated for the report.

Evidence captured

Source evidence consisted of the completed Local AI Lab comparison table, per-run latency/output-rate/token-count panels for all three models, the three representative median-latency outputs, the selected five-run setting, and a separately timed 3:10 complete-suite duration.